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MILITARY SPECIFICATION

THERMOMETER, CLINICAL, VETERINARY

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers a clinical thermometer, self-registering, mercury-in-glass type, for use in veterinary practice in measuring body temperatures of animals.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

PPP-B-566 — Boxes, Folding, Paperboard.

PPP-B-621 — Boxes, Wood, Nailed and Lock-Corner.

PPP-B-636 — Boxes, Fiber.

PPP-B-676 — Boxes, Set-Up, Paperboard.

MILITARY

MIL-L-10547 — Liners, Case, Waterproof.

STANDARDS

FEDERAL

Fed. Test Method Std. No. 626 — Glassware; Methods for Sampling and Testing.

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Stems.* The stems shall be of lens front thermometer tubing having white enameled backing. The tubing shall be Corning Normal standard thermometer lead glass, or equal, having a softening point of 648° Centigrade (C.) ± 5° C., and a coefficient of expansion of 85 ± 2 x 10^{-7} centimeters/centimeters/°C. Tests shall be conducted as specified in 4.4.8.

3.1.2 *Bulbs.* The bulbs shall be made of Corning Normal glass, or equal, having a softening point of 702° C. ± 5° C., and a coefficient of expansion of 86 ± 2 x 10^{-7} centimeters/centimeters/°C. Colored bulbs or bulbs blown from the material of the stem shall not be used. Tests shall be conducted as specified in 4.4.8.

FSC 6535

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3.1.3 *Mercury.* The mercury shall be pure, clean, dry, and free from entrapped gas and foreign material. Thermometers determined to contain entrapped gas when tested as specified in 4.4.3 shall be considered defective.

3.1.4 *Pigment.* The pigment used for filling marks shall be a permanent color complying with the test as specified in 4.4.2. As a result of the test, the pigment shall not fade or discolor in such a manner as to impair legibility. The loss or absence of pigment from one complete graduation or numeral, or the equivalent, shall be cause for rejecting the thermometer.

3.1.5 *Cases.* The cases shall be made of aluminum, nickel—or chromium-plated metal, hard rubber, or suitable plastic material. The barrel and cap of the case shall be of the "screw-on" type.

3.2 **Construction.** Thermometers shall be constructed with a strong glass loop or ring at the distal end, and shall be graduated with the Fahrenheit scale.

3.2.1 *Stem.* The stem shall have white enameled backing in the glass with magnifying lens front, and shall be free from any cracks or other defects. The thickness of the stem shall be not less than 5 nor more than 7 millimeters (mm.) in diameter.

3.2.2 *Bulb.* The bulb shall be of the stubby type, approximately the diameter of the stem.

3.2.3 *Length.* The overall length shall be not less than 125 mm. nor more than 140 mm., measured from the top of the ring to the tip of the bulb.

3.2.4 *Cases.* Unless otherwise specified, each thermometer shall be packed individually in a case (see 3.1.5 and 6.1) and shall be properly protected by cushioning material inside the case. Cases and cushioning material shall be of such size as to be suitable for the thermometers. Cushioning material should not pack or be affected by alcohol which may be used to clean the thermometers.

3.2.5 *Scale.* The scale shall be clearly etched in the stem and filled with a red or black pigment, which shall be resistant to the ordinary methods of disinfecting clinical thermometers. The scale shall cover a minimum range of 94 to 110 degrees Fahrenheit (F.). The highest degree mark shall be not less than ½ inch from the top ring, and the distance from the lowest degree mark to the contraction in the thermometer shall be not less than ¼ inch. There shall be not more than 7 degrees to the inch. The markings shall be at intervals of two-tenths degree on the Fahrenheit scale; each 1-degree line shall be longer than the intervening lines. Numerals on the scale shall be at every 2 degrees F., beginning at the lowest even degree mark. The graduation marks shall be substantially straight, regularly spaced, of uniform width, and perpendicular to the axis of the stem. The width of the graduation marks shall not be greater than 0.012 inch nor less than 0.004 inch, but the width shall in no case be more than the distance between graduation marks. Numerals shall be so placed that the marks to which they pertain can be readily identified.

3.2.5.1 *Accuracy.* At the 98° and 102° F. points, a thermometer shall be accurate to within 0.20° F., and at the 106° point, within 0.30° when tested as specified in 4.4.5.

3.2.5.2 *Consistency of reading.* The two readings of the thermometer at 98° and 102° F. shall not differ from each other by more than 0.20° F., and the two readings at 106° F. shall not differ from each other by more than 0.30° F., when tested as specified in 4.4.5.

3.3 **Time of registration.** Thermometers shall within 2 minutes register the actual body temperature of the animal. Thermometers shall be tested as specified in 4.4.7.

3.4 **Aging.** All thermometers shall be aged by either natural or artificial means so that the performance requirements involved will be met after the thermometer has been introduced into commerce. The natural method

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involves holding the thermometer at room temperature for a period of 4 months after completion of the constriction. The artificial method involves heat treatment of bulb and constriction at a temperature of at least 840° F. for at least 3 hours, followed by cooling to 500° F. at a rate not to exceed 100° per hour.

3.5 **Hard shakers.** Thermometers shall be so constructed that the column of mercury can be shaken down by hand without undue effort, when tested as specified in 4.4.4.

3.6 **Nonretreating index.** Thermometers shall be made so as to have a nonretreating index, when tested as specified in 4.4.6.

3.7 **Identification marking.** Each thermometer shall bear in permanent legible characters, filled with a pigment, the name or registered trade-mark of the manufacturer, and serial number. Marking shall be of such size and location as not to interfere with the reading of the thermometer.

3.8 **Workmanship.** Workmanship shall be first class throughout. Thermometers shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 The provisions contained in this section are applicable to inspection performed by and for the Government. Sampling, inspection, and test procedures for components or materials described shall be in accordance with this specification and with specifications and standards referenced herein to the extent applicable, except that this specification shall govern in the event of conflict.

4.1.1 *Supplier responsibilities for inspection.* Such examinations and tests as are set forth in this specification or as shall otherwise be appropriate or necessary to insure that supplies conform to specification requirements, shall be performed by and at the expense of the supplier. Suppliers who do not have facilities adequate for such tests shall arrange for the use of test facilities acceptable to the Government. Records of examinations and tests performed by the supplier shall be maintained by the supplier and made available to the Government, upon the Government's request, at any time, or from time to time, for a period of 2 years after delivery of the supplies to which such records relate.

4.1.2 Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc. investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 *Certificates of quality.* When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained may be furnished in lieu of actual performance of such testing by the contractor.

4.2 **Sampling.** Sampling shall be conducted in accordance with Standard MIL-STD-105 and appendix thereto.

	Inspection level	*AQL (percent defective)*
For visual examination		
Major defects	II	1.0
Total (major and minor combined).	II	4.0
For dimensional examination	L8	2.5
For end item testing		
Permanency of color markings	L8	1.0
Other tests	II	1.0

4.2.1 *For raw materials.* Samples of the glass used for the stems and bulbs in the construction of the thermometers, prepared in accordance with Fed. Test Method Standard No. 626, shall be selected for tests (see 4.4.8).

4.2.2 Where possible, the same sample shall be used for the determination of two or more tests.

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4.3 Inspection.

4.3.1 *Materials.* Prior to fabrication, all component materials of the thermometers shall be examined and tested in accordance with specification requirements.

4.3.2 *End item.* The thermometers shall be inspected to determine compliance with all requirements contained in sections 3 and 5 of this specification. Nonconformance with these requirements will be permitted to the extent indicated in 4.2.

4.3.3 *Classification of defects.* Inspection shall be conducted in accordance with the following classification of defects:

Major	*Categories and defects**
101.	Crack or chipped out area in stem or bulb.
102.	Case cracked or broken.
103.	White enameled backing missing or not complete.
104.	Magnifying lens front does not function properly, or omitted.
105.	Not graduated in Fahrenheit scale.
106.	Graduation markings not in intervals of two-tenths degree.
107.	Numerals not marked at every 2 degrees Fahrenheit beginning at lowest even degree.
108.	Graduation markings not properly etched, or missing.
109.	Scale range not 94° through 110°F.
110.	Thermometer not free of sharp edges or projections.
111.	Thermometer not properly protected by cushioning material in case.
112.	Thermometer not free of breaks or foreign matter in column.
113.	Barrel and cap of case not "screw-on" type.
114.	Identification marking interferes with reading of the thermometer.
115.	Cases and cushioning material not of size suitable for the thermometers.
Minor	
201	Loop or ring missing at distal end.
202.	Etchings not in red or black pigment.
203.	Degree lines not longer than the intervening lines.
204.	Graduation marks not uniform and regularly spaced.
205.	Numerals so placed that marks to which they pertain are not readily identified.
206.	Identification markings missing or not legible.
207.	Poor workmanship not resulting in a major defect.

* Inspection is not restricted to the classified possible defects listed above.

4.3.4 *Dimensional examination.* The thermometers shall be inspected for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted in accordance with the applicable specifications, as necessary, to determine compliance with this specification. In the absence of detailed descriptions of test methods in these specifications, compliance with the requirements shall be determined by methods appropriate to the several requirements.

4.4.1 *Aging.* Thermometers shall be aged by either natural or artificial means so that the performance requirements involved will be met after the thermometers are introduced into commerce. The natural method involves holding at room temperature for a period of 4 months after completion of the constriction. The artificial method involves heat treatment of the bulb and constriction at a temperature of at least 840° F. for at least 3 hours, followed by cooling to 500° F. at a rate not to exceed 100° F. per hour. In the absence of evidence satisfactory to the procuring activity that the requirements outlined above have been met, the thermometers may be held for a period of not more than 3 months after delivery before testing.

4.4.2 *Test for permanence of color markings.* Sample thermometers shall be subjected to the following tests. During the tests, the solutions shall be maintained at a temperature between 70° and 90° F. Removal periods shall be approximately of 15 minutes duration, or as long as required to dry a thermometer. The test schedule shall be of alternate immersion in the solution specified for periods of one hour and drying the thermometers by wiping with soft tissue or a turkish towel. Five such complete schedules of immersion and drying shall be completed on each of two successive days. The thermometers shall remain in the solution in the intervening night. At the conclusion of each test, the markings on each thermometer shall meet the specified requirements in 3.1.4.

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4.4.2.1 *Test "A".* Sample thermometers shall be totally immersed in a solution consisting of one part of tincture of green soap, N.F., and four parts of water.

4.4.2.2 *Test "B".* Sample thermometers shall be totally immersed in a 5 percent by weight phenol solution.

4.4.3 *Test for entrapped gas.*

4.4.3.1 Thermometers in which visual inspection shows the presence of gas in the bulb shall be classified as defective.

4.4.3.2 All thermometers in which there is sufficient gas or other foreign material to cause separation of the mercury column above the constriction during shipment, test, or use, shall be classified as defective. To test for this condition, the thermometer shall be heated so that the reading is 102° F. or 39° C. The mercury shall be shaken away from the bulb in the centrifuge at 200 revolutions per minute with the top of the thermometer 32 centimeters (cm.) from the axis of rotation. A broken mercury column above the constriction shall be an indication that there is gas or other foreign material present.

4.4.4 *Hard shakers.* The index shall fall below 96° F. when the thermometer is mounted in a centrifuge with the end of the bulb 17 cm. from the axis of rotation, and the centrifuge is brought up to a speed of 580 revolutions per minute.

4.4.5 *Tests for accuracy and consistency of reading.*

4.4.5.1 Thermometers shall be compared at 98°, 102°, and 106° F., with a thermometer certified by the National Bureau of Standards for this purpose, by heating in a constantly-stirred water bath, removing from bath, and reading. The temperature of the bath shall be raised at a substantially uniform rate during the last 2 minutes before the test temperature is reached, and the rise during this 2-minute period shall be not more than 0.2° F. The thermometer shall be removed from the bath as soon as the temperature of test is reached, except in the second test at 106° F. where the procedure specified under 4.4.5.2 shall be followed. The reading shall be observed after removal from the bath. A suitable magnifying glass shall be used and the readings estimated to the nearest one-tenth division (graduation interval), or better. Two independent comparisons shall be made and recorded on each thermometer at each test temperature.

4.4.5.2 In the first test, at 106° F., the thermometers shall be removed from the bath as soon as the test temperature is reached. In the second test, at 106° F., the bath shall be brought up to the test temperature and then allowed to cool slowly to 105° F., or below, at a uniform rate not exceeding 1° F. in 3 minutes. The thermometers shall then be removed from the bath and read, and the readings recorded as before.

4.4.5.3 *Accuracy.* No individual reading, in comparison with the certified standard, shall be in error by more than the amount permitted in 3.2.5.1.

4.4.5.4 *Consistency.* The two readings at each test point shall not differ from each other by more than the amount permitted in 3.2.5.2.

4.4.6 *Nonretreating index.* After removal from a constantly-stirred water bath in which the thermometer is brought to a temperature of not less than 104° F., nor more than 108° F., the thermometer shall retain its registration until manually changed.

4.4.7 *Time of registration.* A constantly-stirred water bath shall be brought to a constant temperature of 100° F. The thermometers shall then be placed in the bath for a period of not more than 2 minutes, then withdrawn from the water bath and read. The thermometers shall be capable of registering the actual temperature of the water within the 2-minute period.

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4.4.8 *Physical properties of glass.* Tests for physical properties of glass shall be in accordance with the following methods of Fed. Test Method Standard No. 626:

Coefficient of expansion	Method No. 240 or 241
Softening point	Method No. 201

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 *Unit of issue.* One (1) thermometer, as specified, constitutes one unit.

5.1.2 *Packaging quantities.* The number of units of issue indicated in table I shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in procurement document.

TABLE I. *Packaging quantities*

Federal stock number	Unit package	Intermediate package	Exterior container
6535–812–7000	1 unit	24 units	1,152 units

5.1.2.1 *Packing variation permitted.* If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.3 *Level A.*

5.1.3.1 *Unit package.* Each unit shall be packaged in a carton of appropriate size constructed in accordance with Specification PPP–B–566 or PPP–B–676. Each carton shall be adequately secured to prevent accidental opening.

5.1.3.2 *Intermediate package.* Intermediate package shall be a carton of appropriate size constructed in accordance with Specification PPP–B–636. Each carton shall be adequately secured to prevent accidental opening.

5.1.4 *Level C.* Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.2 Packing.

5.2.1 *Level A.*

5.2.1.1 *Exterior container.* Exterior container shall be a box constructed in accordance with Specification PPP–B–621, table IV.

5.2.1.2 *Case liner.* Each level A box shall be lined with a waterproof case liner conforming to Specification MIL–L–10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.2.1.3 *Strapping.* Strapping, when required, shall be in accordance with applicable box specification.

Note: Strapping will not be required for shipment forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 *Level B.*

5.2.2.1 *Exterior container.* Exterior container shall be a box constructed in accordance with Specification PPP–B–621, table II.

5.2.3 *Level C.* The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 *Interior packages.* Each unit package and each intermediate package shall be marked in accordance with section 5 of Standard MIL–STD–129.

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5.3.2 *Exterior container.* Exterior container shall be marked in accordance with Standard MIL-STD-129, except that 6.2(c), 6.2(d), and 6.5 shall not be applicable. Service color and service symbol markings are required only on direct shipments from the contractor's plant to overseas activities.

6. NOTES

6.1 **Ordering data.** Procurement documents should specify the following:

(a) Number and date of this specification.

(b) When case is not required with thermometer (see 3.2.4).

(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification covers the following item appearing in the Armed Services Medical Stock List:

6535-812-7000
THERMOMETER, CLINICAL, VETERINARY.

6.3 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Military requirements.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:
Army—Army Medical Service
Navy—Bureau of Medicine and Surgery
Air Force

Preparing activity:
Navy—Bureau of Medicine and Surgery (MMSA)

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